

ECE448

Spring 2025

Lesson 21

DTMF



UNITED STATES
AIR FORCE
ACADEMY

NOMINAL SCHEDULE

21- DTMF
22- DTMF

Lesson	Topics	Notes
22	RF Reverse Engineering	Final Project Proposal due
23	RF Reverse Engineering	
24	RF Reverse Engineering	
25	Flex day	RF RE Project Due
26	ADS-B	
27	Project Work Day	Background Research due
28	ADS-B	
29	ADS-B	
30	AIS	
31	AIS	Design Description due
32	WBFM Radio	ADS-B/AIS Project
33	HD Radio	
34	HD Radio	
35	Project Work Day	Prototype Description due
36	QAM	
37	QAM /OFDM	
38	QAM /OFDM	
39	Project Day	
40	Final Project Presentations	Final Project due

LESSON ~~25~~ 21 LEARNING OBJECTIVES

- 1.1 Build OOT blocks using both Conda (Windows) and Ubuntu
- 1.2 Describe what DTMF is and how it transmits information.
- 1.3 Build a DTMF encoder and transmitter.
- 1.4 Describe how the Goertzel algorithm can be used to calculate
- 1.5 Build a DTMF receiver and decoder.

BUILDING CUSTOM BLOCKS

Conda

- Install VSCode Enterprise 2019
 - Enable MSVC142 components
 - The download should be something like 11GB

Linux:
• `mkdir build`
• `cd build`
• `cmake ..`
• `make -j4`
• `sudo make install`

- Install dependencies:

```
[conda install gnuradio-build-deps  
cxx-compiler c-compile]
```

- Enter the "build" directory and execute:

```
[cmake -G Ninja \  
-DCMAKE_BUILD_TYPE=Release \  
-DCMAKE_TRY_COMPILE_TARGET_TYPE=STATIC_LIBRARY \  
-DCMAKE_INSTALL_PREFIX="%CONDA_PREFIX%\Library" \  
-DCMAKE_PREFIX_PATH="%CONDA_PREFIX%\Library" \  
-DGR_PYTHON_DIR="%CONDA_PREFIX%\Lib\site-packages" ..]
```

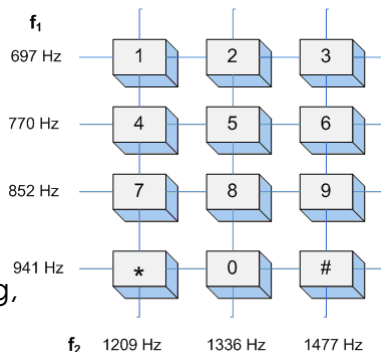
```
[cmake --build .]
```

```
[cmake --build . --target=install]
```

```
conda activate environment
```

DUAL TONE MULTIPLE FREQUENCY

- Each tone pair (one high, one low) uniquely represents a digit
- [Reference Info](#)
- Tone length: 40ms and up
- Space length: 20ms and up
- Decoding software: multimon-ng, DTMF (android), etc.
- [Not just for telephones!](#)



BUILDING CUSTOM BLOCKS IN CONDA

Demo

- Android DTMF
- multimon-ng
- How else could we decode it?

Tasks

- Build a GNURadio DTMF transmitter (easy)
- Build a decoder (not so easy)

GOERTZEL

DTMF Signal for Key 6

wirelesspi.com

